

Ecological and histological aspects of tail loss in spiny mice (Rodentia: Muridae, *Acomys*) with a review of its occurrence in rodents

Eyal Shargal¹, Lea Rath-Wolfson², Noga Kronfeld¹ and Tamar Dayan^{1*}

¹Department of Zoology, Tel Aviv University, Ramat Aviv, Tel Aviv 69978, Israel

²Rabin Medical Center, Golda Campus, Petah Tiqva, Israel

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Abstract

Many individual nocturnal common spiny mice *Acomys cahirinus* and diurnal golden spiny mice *A. russatus* in the field are found missing all or part of their tails. Possibly, this is a predator avoidance mechanism. A 25-month field study revealed that at Ein Gedi (Israel) percentage tail-loss is 63% in male and 44% in female golden spiny mice, and 12% in male and 25% in female common spiny mice. Tail loss is significantly more common in golden spiny mice than in common spiny mice, possibly reflecting differences in predation risk or predator efficiency between the different microhabitats used by these species and in their different activity times. However, inter- and intraspecific aggressive interactions may also account in part for this pattern. Few significant differences in longevity, body mass, or reproductive condition were found between tailed and tail-less spiny mice, suggesting an advantage to tail-less individuals. Histological sections revealed a plane separating the skin layer from the underlying muscles and vertebrae, facilitating loss of the skin with little bleeding. The remainder (muscles and bone) is later chewed off by the mouse. A survey of published cases of tail loss in rodents revealed that this phenomenon occurs in at least 35 species and has evolved separately in eight rodent families, with no clear pattern in systematics, geography, or habitat use.

Key words: spiny mice, tail loss, autotomy, histology, predation

INTRODUCTION

Autotomy, the loss of a body part, sometimes accompanied by subsequent regeneration, is a common feature of many invertebrates and vertebrates. It is often considered an anti-predatory defence mechanism (e.g., Robinson, Abele & Robinson, 1970; Arnold, 1984; Cooper & Vitt, 1985; Formanowicz, Brodie & Bradley, 1990), in which body parts that are attacked by a predator may be sacrificed in order to save the individual animal. This mechanism has apparently evolved independently in disparate taxa such as spiders (e.g. Klawinski & Formanowicz, 1994; Cloudsley-Thompson, 1995), insects (e.g. Carlberg, 1981, 1984; Rietschel, 1987), echinoderms (e.g. Lawrence, 1992), crustaceans (e.g. Hirvonen, 1992; Abello *et al.*, 1994; Juanes & Smith, 1995), slugs (e.g. Deyrup-Olsen, Martin & Paine, 1986; Pakarinen, 1994), reptiles (e.g. Arnold, 1988), and amphibians (e.g. Labanick, 1984). The sacrificed organs include chelipeds (e.g. Robinson *et al.*, 1970; Juanes & Smith, 1995), walking legs

(Rathmayer & Bevenegut, 1986; Formanowicz, 1990), arms (Oji & Okamoto, 1994) and tails (Feder & Arnold, 1982; Ducey, Brodie & Barness, 1993).

Among vertebrates, caudal autotomy, the ability to shed the tail (as defined by Arnold, 1988; see also Roth & Roth, 1984 for alternative definitions) in reptiles and amphibians is a common and well recognized anti-predator defence mechanism and has been studied at both the anatomical–histological and ecological–evolutionary levels (e.g. Arnold, 1984; Abdel-Karim, 1993; Salvador *et al.*, 1996). In mammals, accounts of tail loss among rodents have appeared sporadically in the literature throughout this century and have usually been related to predator avoidance (e.g. Cuenot, 1907; Gogl, 1930; Layne, 1972; see also Heneberg, 1910). Early research dealing with the tail histology of these species revealed interspecific differences in the mechanism of tail loss (Mohr, 1941). Generally, studies of tail loss in rodents were not accompanied by field research and reports of the extent of this phenomenon in natural populations remain largely anecdotal.

We studied tail loss in two sympatric congeneric species of spiny mice: the common spiny mouse *Acomys cahirinus* and the golden spiny mouse *A. russatus*. The

*All correspondence to: T. Dayan, Department of Zoology, Tel Aviv University, Ramat Aviv, Tel Aviv 69978, Israel

two species co-occur in areas of rocky deserts in the south of Israel and are active at different times: *A. cahirinus* is nocturnal while *A. russatus* is diurnally active (e.g. Kronfeld *et al.*, 1994), apparently as a result of interspecific competition with *A. cahirinus* (Shkolnik, 1971). We encountered many individuals of the two species that had lost their tail both in the field and in our captive colony.

Our field and laboratory study of tail loss in spiny mice set out to: look at the histological mechanism of tail loss; determine the extent of this phenomenon in free-living populations; assess possible associated costs in longevity, body mass, and reproduction; gain insight into the possible ecological-evolutionary basis for this phenomenon.

Here we report the results of our study. We also survey the literature on tail autotomy in rodents in an attempt to find an ecological and/or taxonomic pattern for this phenomenon.

MATERIALS AND METHODS

We studied histological sections of tails of both species of spiny mice, and for comparison of the laboratory house mouse (*Mus musculus*). We examined intact tails, tails that had just lost their skin sheath (mice were sacrificed within 2 min of skin sheath loss), and tails that had been partially lost some time ago. Tails were routinely fixed in buffered formalin 10% and then embedded in paraffin. We took longitudinal sections stained with H&E (haematoxylin and eosin).

We trapped common spiny mice and golden spiny mice for 25 consecutive months in a 5000 m² grid near Kibbutz Ein-Gedi, Israel, near the Dead Sea (31°28'N, 35°23'E, 100–350 m below sea level), using 200 Sherman collapsible traps during the first year and 100 traps subsequently. We trapped each month for 3 consecutive days and nights, checking the traps at dawn and dusk and at 4 equal intervals during the day. We recorded the species, sex, reproductive condition, and condition of the tail of each individual trapped, and marked individuals by toe-clipping.†

We compared the percentage tail loss of the different sexes and species using 2-tailed chi-square tests. In order to study a possible cost in longevity associated with tail loss, we compared the length of time during which individual mice were repeatedly captured (months between first and last record) for individuals with tails and those without part or all of their tail, using a *t*-test (after ascertaining normality of distributions using the Shapiro–Wilk's *W*-test). Only individuals that were trapped for more than 1 month, and therefore deemed to be residents, were included in the analysis, thus excluding transients. We did not use the Jolly-Seber

probability of survival estimates because of the low population densities, which would render the results inaccurate. The limited trapping period (25 months) meant that estimated longevity of some individuals was artificially reduced. This bias should affect tailed and tail-less mice to the same degree, but may result in a reduced difference between the 2 groups.

To study a possible cost in reproduction associated with tail loss, we also compared the reproductive condition (pregnancy or evident nipples for females and scrotal testes for males) of mice with and without tails, using a 2-tailed chi-square test, during the period of reproduction (February–August; Shargal, 1997). Regarding a possible cost in physical condition, we compared the mean body mass of mice with and without tails using a Student's *t*-test (excluding immature individuals).

RESULTS

When one gently holds the tail of an individual spiny mouse, the animal immediately walks away – the skin of the tail separates easily from the underlying muscles and vertebral column, with only slight bleeding, and breaks, leaving only the sheath (consisting of epidermis and upper part of the dermis only) in one's grasp. There is no need to exert pressure or to pull. The remainder of the tail, bone and tissue, is later consumed by the mouse, which is left with a considerably shorter tail or no tail at all.

Longitudinal histological sections revealed the presence of a longitudinal plane between the skin and the underlying muscles and vertebrae (Fig. 1). This plane, which is formed by a separation in the connective tissue fibres that bind the skin to the underlying tissues, enables the skin to separate with ease (Fig. 2). This phenomenon was found in both species. In the tail of the laboratory mouse there is no such plane, and the skin is fully connected with connective tissue to the underlying layers (Fig. 3).

The occurrence of this separating plane appears to facilitate the loss of the skin sheath and of the ensuing tail. However, in those tails that were only partially lost, we observed fibrosis in the remaining stump. This part could then no longer disconnect with ease when tugged. We were unable to discern any specific points where the skin is most likely to break. However, connective tissue fibres do occur sporadically along the plane and may be the points where the skin tears. This histology may account for the degree of variation in tail loss encountered in the field (see below).

In the 25-month field study we found a high percentage of individuals with partial or total tail loss: 17 out of 27 (c. 63%) male *A. russatus*, seven out of 16 (c. 44%) female *A. russatus*, three out of 26 (c. 12%) of male *A. cahirinus*, and four out of 16 (25%) female *A. cahirinus*. The difference in percentage tail loss between the two species is statistically significant

† Editor's note: The Ethical Committee of the Zoological Society of London considers that toe-clipping is no longer acceptable as a routine procedure for marking animals.

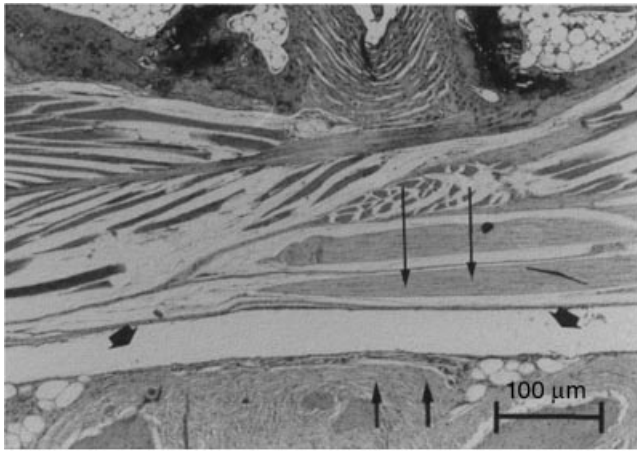


Fig. 1. Longitudinal histological section of a common spiny mouse *Acomys cahirinus* tail. The golden spiny mouse *A. russatus* tail is identical in histological detail. Note the cleavage plane (thick arrows) between the muscles (long thin arrows) and skin and subcutaneous tissues (short thin arrows).

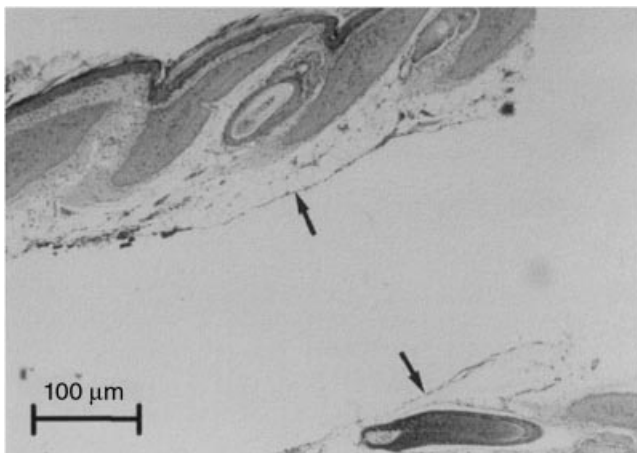


Fig. 2. The separated tail skin of a common spiny mouse *Acomys cahirinus* tail showing the epidermis and subcutaneous fat (arrows). Note the absence of the muscular layer and vertebrae in the middle.

($P=0.0004$), while the differences between males and females within each species are not statistically significant.

Tail loss ranged from partial (in most individuals) to full. Two *A. russatus* individuals initially recorded with partial tail loss eventually lost the remainder of their tails.

There was no significant difference in length of time of repeated captures between individuals with and without tails, when we compared males to males or females to females of each species (Table 1). However, when we compared mixed-sex samples with and without tails there was a significant difference between common spiny mice, with the group that experienced tail loss demonstrating significantly greater longevity ($P=0.044$). No significant difference was found for mixed-sex golden spiny mice.

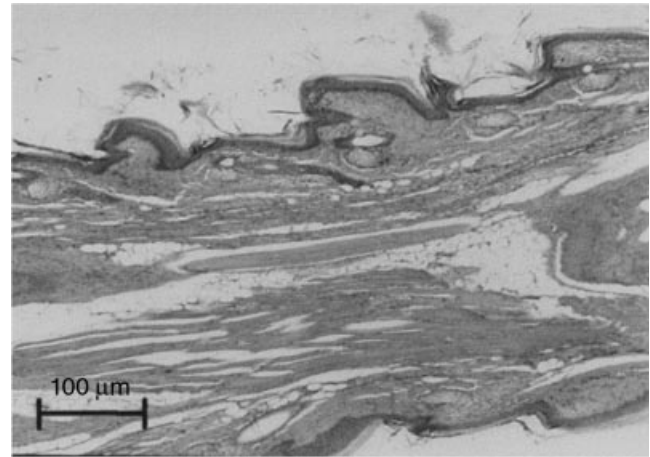


Fig. 3. Longitudinal histological section of a laboratory mouse *Mus musculus* tail. There is no cleavage plane in the histological section.

There was no significant difference in reproductive condition between tailed and tail-less female common spiny mice or between mixed-sex common spiny mice (there was only one tail-less male, so a comparison among males was not carried out) (Table 1). Among golden spiny mice, however, while there was no significant difference between tailed and tail-less females, we found a significant difference between males – a greater percentage of tail-less males were found in reproductive condition ($P=0.031$), and the same was also true for mixed-sex groups ($P=0.031$).

There was no significant difference in body mass between tailed and tail-less golden spiny mice (we compared males to males and females to females but did not compare mixed-sex samples since there are significant differences in body mass between the sexes; Shargal, 1997) (Table 1). Among common spiny mice we found no significant difference between males, but the difference between females was significant ($P=0.0115$), with the mean body mass of tail-less females significantly greater.

DISCUSSION

Our study demonstrates that tail loss is very common among natural populations of spiny mice at Ein-Gedi and that there is a distinct underlying mechanism. Earlier literature on tail loss in rodents sometimes dealt with the histological mechanisms involved but seldom addressed the extent of this phenomenon in populations in the wild (Table 2), so comparisons are limited. From the data available it appears that the occurrence of tail loss in other species of rodents is usually much less frequent (Table 2), and that tail loss in spiny mice is indeed an unusually common phenomenon.

Because longevity, body mass, and reproductive condition may well be related (but with unknown correlations), we cannot combine our statistical tests of

Table 1. Average length of time of repeated captures, mean body mass, and percentage of individuals in reproductive condition for tailed (+) and tail-less (–) golden spiny mice (*A.r.*) and common spiny mice (*A.c.*), in males (M), females (F), and mixed-sex samples (M+F).

Species and sex	Tail	Repeated captures (months)			Body mass (g)			Percentage in reproductive condition	
		Mean	S.D.	<i>n</i>	Mean	S.D.	<i>n</i>	%	<i>n</i>
<i>A.r.</i> M	+	8.0	5.9	7	42.9	5.6	10	7.6	13
<i>A.r.</i> M	–	10.5	6.7	15	43.0	6.1	16	46.6	15
<i>A.r.</i> F	+	6.1	4.4	6	38.6	6.4	10	38.4	13
<i>A.r.</i> F	–	7.5	5.3	6	41.5	8.3	3	80.0	5
<i>A.r.</i> M+F	+	7.1	5.1	13	40.8	6.3	20	23.0	26
<i>A.r.</i> M+F	–	9.6	6.3	21	42.8	6.3	19	55.0	20
<i>A.c.</i> M	+	5.0	2.5	17	34.7	6.0	22	38.4	26
<i>A.c.</i> M	–	7.7	1.5	3	34.2	1.0	2	100	1
<i>A.c.</i> F	+	7.1	5.8	7	28.0	6.8	7	46.6	15
<i>A.c.</i> F	–	12	9.2	3	36.8	1.6	2	33.0	3
<i>A.c.</i> M+F	+	5.7	3.8	24	33.0	6.7	29	41.4	41
<i>A.c.</i> M+F	–	9.8	6.3	6	35.5	1.9	4	50.0	4

Table 2. Published cases of rodent species that undergo tail loss

Species and family	Reference
<i>Funisciurus substriatus</i> (Sciuridae)	Robinson, 1967
<i>Tamias striatus</i> (Sciuridae)	Layne, 1972
<i>Liomys pictus</i> (Heteromyidae)	Mendoza pers. comm.
<i>Perognathus fallax fallax</i> (Heteromyidae)*	Sumner & Collins, 1918
<i>Perognathus panamintinus bangsi</i> (Heteromyidae)*	Sumner & Collins, 1918
<i>Neotoma lepida</i> (Muridae)	Gros pers. comm.
<i>Peromyscus boylii</i> (Muridae)	Nowak, 1991; Sumner & Collins, 1918
<i>Peromyscus floridanus</i> (Muridae)	Layne, 1972
<i>Sigmodon hispidus</i> (Muridae)	Dunaway & Kaye, 1961
<i>Apodemus sylvaticus</i> (<i>Mus sylvaticus</i>) (Muridae)	Cuenot, 1907; Mohr, 1941
<i>Apodemus flavicollis</i> (Muridae)	Mohr, 1941
<i>Apodemus agrarius</i> (Muridae)	Mohr, 1941
<i>Rattus norvegicus</i> (Muridae)	Mohr, 1941
<i>Rattus rattus</i> (Muridae)	Mohr, 1941
<i>Zyomys woodwardi</i> (Muridae)	Ride, 1970
<i>Zyomys pedunculatus</i> (Muridae)	Ride, 1970
<i>Lophuromys</i> sp. (Muridae)	Anonymous
<i>Acomys russatus</i> (Muridae)	This paper
<i>Acomys cahirinus</i> (Muridae)	This paper
<i>Acomys wilsoni</i> (Muridae)	Takata pers. comm.
<i>Acomys percivali</i> (Muridae)	Takata pers. comm.
<i>Mus musculus</i> L. (Muridae)	Mohr, 1941
<i>Glis glis</i> (Gliridae)	Mohr, 1941
Dormice (Gliridae) no details on species	Robinson, 1967
<i>Eliomys quercinus</i> (Gliridae)	Cuenot, 1907; Gogl, 1930; Mohr, 1941
<i>Muscardinus avellanarius</i> (Gliridae)	Gogl, 1930; Mohr, 1941
<i>Dryomys nitedula</i> (Gliridae)	Mohr, 1941
<i>Graphiurus crassii caudatus</i> (Gliridae)	Thomas, 1905 in Mohr, 1941
<i>Graphiurus</i> sp. (Gliridae)	Thomas, 1905 in Mohr, 1941
<i>Lagidium peruanum</i> (Chinchillidae)	Pearson, 1948
<i>Mysateles melanurus</i> (<i>Capromys melanurus</i>) (Capromyidae)*	Mohr, 1941
<i>Mysateles nana</i> (<i>Capromys nanus</i>) (Capromyidae)*	Mohr, 1941
<i>Mysateles prehensilis</i> (Capromyidae)	Mohr, 1941
<i>Octodon degus</i> (Octodontidae)*	Fulk, 1975; Nowak, 1991; Woods & Boraker, 1975
<i>Proechimys</i> sp. (and other Echimyidae)*	Allen, 1918; Walker, 1968
<i>Proechimys longicaudatus</i> (Echimyidae)	Redford & Eisenberg, 1992

* Species with tail breaks across a vertebrae.

these characters in tailed and tail-less spiny mice. We can only say that most of the differences between groups were not significant, and that the few significant differences that were observed suggest an advantage in longevity, body mass, and reproductive condition to tail-less spiny mice over those with tails. If these traits are taken as indices of fitness, it would appear that there is no cost to be paid for tail loss. Tail loss is accompanied by energetic, social and locomotory costs in lizards (Fox & Rostker, 1982; Fox, Heger & Delay, 1990; Cooper & Vitt, 1991; Wilson, 1992; Martin & Salvador, 1993; Kaiser & Mushinsky, 1994; Salvador, Martin & Lopez, 1995; but see Daniels, 1983) and salamanders (e.g. Maiorana, 1977). Among spiny mice, however, tails do not serve for energy storage, and although they may have some locomotory function, we can point to no such clear costs. It has previously been argued that species in which the tail loss involves low costs, may lose them more readily (Arnold, 1988).

Arnold (1988: 241) suggested that 'the probability that an individual lizard will have lost its tail necessarily increases with time, although the increase is often not uniform, and young animals are frequently more prone to caudal autotomy than are adults'. The same should be true for spiny mice, so it is surprising that the individuals we first trapped with or without tails were found to enjoy the same life expectancy; because tail-less spiny mice could be expected to be older on average, their life expectancy could be expected to be lower. However, it is possible that tail loss occurs primarily in young individuals. Spiny mice are fairly long-lived rodents (we trapped the same individuals for 2 consecutive years), which achieve adult size and mass within a few weeks. Unless we trapped individuals in the first few weeks of life, we would be unable to detect a tendency to juvenile tail loss. Alternatively, it is possible that mortality in spiny mice is not age-dependent.

What causes tail loss in spiny mice? Predation pressure, a strong ecological, and evolutionary force, has often been invoked to explain caudal autotomy. Arnold (1988: 238) suggests that 'the frequent efficacy of autotomy in allowing escape is well known to anyone who has tried to catch lizards' and some experimental research using various predators supports this statement (Congdon, Vitt & King, 1974; Daniels, 1983; Daniels, Flaherty & Simbotwe, 1986). In our field experience, tail loss is an effective mechanism of escape from the human grasp.

Spiny mice suffer predation from Blandford's foxes *Vulpes cana* (Geffen *et al.*, 1992), snakes, Hume's tawny owls *Strix butleri* (Mandelik, pers. obs.), and possibly kestrels *Falco tinnunculus*, and they have evolved certain defence mechanisms as a response to this pressure – spines on the rump, which are particularly tough in the golden spiny mice, may deter predators; and immunity to snake venom (Weissenberg, Bouskila & Dayan, 1996). Tail autotomy may well be a further mechanism. Moreover, the skin of spiny mice is easily torn and heals rapidly. In other taxa this phenomenon has been suggested as an additional anti-predatory defence mechanism (Bauer & Russell, 1992).

Spiny mouse tails do not regenerate, so if their loss is an anti-predatory escape tactic, then it allows only a single escape. If this mechanism reduces predation then one should expect mice with tails to be less prone to being preyed upon than those without, as has been demonstrated for lizards (Congdon *et al.*, 1974; Daniels, 1983; Daniels *et al.*, 1986). Our sample sizes may be insufficient to detect such a difference, but it is also not inconceivable that spiny mouse tails, which add 55–85% to their body lengths (data from Mendelssohn & Yom-Tov, 1987), simply make them more prone to predation.

In lizards the tail is often used as a decoy, drawing the predator's attention to it either by waving or by bright colours, or both (Vitt & Cooper, 1986; Cooper & Vitt, 1991). Spiny mice have simple murid type tails, covered with scales and sparse fur, with no conspicuous colours or tufts of hair. We noted no behavioural adaptations (such as tail waving) associated with them. While lizard tails offer energy returns to the predators, thus inducing them to give up the chase (Arnold, 1988), tail sheaths and even whole tails of rodents offer little to tempt a predator. Moreover, contrary to lizard tails, which continue to move and to attract the predator post-autotomy, the skin sheaths of rodent tails remain limp and lifeless in the predator's grasp. Obviously, the evolution of tail loss in these species is not coupled with the evolution of the tail as an active decoy. Possibly this mechanism merely facilitates discarding an unessential organ when it is grasped by a predator or a congener, thus saving the animal's life.

If tail loss results from predation attempts, the difference in percentage tail loss between the nocturnal common spiny mouse and the diurnal golden spiny mouse may reflect differences in predation pressure caused by interspecific differences in microhabitat use (analyzed by Krasnov *et al.*, 1996; Shargal, 1997) or differences in predation pressure between day and night (see Pianka and Pianka, 1976; Schall & Pianka, 1980, for this phenomenon in lizards; but see Schoener, 1979; Jaksic & Greene, 1984). The fact that, when given the chance, in the absence of their congener, golden spiny mice switch from diurnal to nocturnal activity (Shkolnik, 1971) suggests a cost inherent in diurnal activity for this species. Possibly, this cost, or part of it, is the cost of predation. However, the difference in percentage tail loss may also reflect spatial or temporal variation in predator efficiency (see Medel *et al.*, 1988). We note that the more frequently-lost golden spiny mouse tails are significantly shorter when intact than tails of the common spiny mouse (Student's *t*-test: $P=0.055$ for males, $P<0.001$ for females; data from Mendelssohn & Yom-Tov, 1987), despite no significant interspecific difference in body length ($P=0.29$) for males, and that female golden spiny mice are significantly longer in body length than female common spiny mice ($P=0.0117$). This difference may suggest that long tails could be disadvantageous to spiny mice under heavy risk of predation.

However, forces other than predation may contribute

to tail loss. Tail loss resulting from intraspecific aggression has been reported in salamanders (Jaeger, 1981) and lizards (Arnold, 1984) and has been speculated upon also for rodents (Pearson, 1948; Dunaway & Kaye, 1961). When spiny mice are kept in cages at high densities, intraspecific aggression occurs, resulting in injuries such as torn ears, open wounds, and tail loss. When kept in enclosures, individuals of both species tend to aggregate, indicating some social interaction in the wild (see also Rosenzweig & Abramsky, 1985). Intraspecific aggression may also therefore account, at least in part, for tail loss in the wild.

Shkolnik (1971) suggested that the common spiny mouse excludes the golden spiny mouse from nocturnal activity and forces it into a diurnal activity pattern. However, the precise cues for the exclusion remain to be studied. If aggressive interspecific interference is involved, this may account for the higher incidence of tail loss in the golden spiny mouse.

Two major mechanisms of tail loss in rodents are described in the literature (see Mohr, 1941):

(1) The mechanism found in spiny mice, which involves loss of tail skin and subsequent loss of tail, with the animals apparently chewing off the remainder. This appears to be the more common mechanism and occurs in five families and at least 26 species (see Table 2) other than spiny mice, although the precise underlying histology is not identical in all of them.

(2) Tail-break, with the tail breaking across a vertebrae, similar to the mechanism described for geckos (e.g. Werner, 1965). This is apparently the less common phenomenon and has so far been described in only six species of three different families (see Table 2).

It appears that tail loss in rodents has evolved independently several times. It occurs in rodents of all recognized suborders, in at least eight of 29 families, 23 of 426 genera, and 35 of 1814 species, with no clear pattern in systematics, geography or habitat use. Although future research may add a number of species to our list, it appears that tail loss in rodents is not nearly so common as caudal autotomy in lizards, perhaps because the superior abilities of lizard tails (active shedding of tails, post-autotomous movement and regeneration) render them more effective as an antipredatory mechanism.

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